AI Engineering & Production LLM Applications Guide

Comprehensive Developer Masterclass & Practical Guide

ARTIFICIAL INTELLIG

Author / Publisher: AI Research & Deployment Lab

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Chapter 1: High-Accuracy RAG Architectures

Beyond Naive Semantic Search: Hybrid Reranking & Context Compression

Naive RAG pipelines suffer from context loss and high false-positive retrieval rates when queries are nuanced.

By pairing sparse BM25 keyword matching with dense embedding cosine similarity and running results through a Cross-Encoder Reranker, retrieval precision increases by up to 45%.

```
# Hybrid Rerank Retrieval Pipeline
dense_docs = vector_store.similarity_search(query, k=25)
sparse_docs = bm25_retriever.get_relevant_documents(query)
merged = deduplicate(dense_docs + sparse_docs)
final_top_k = cohere_rerank(query, merged, top_n=5)
```

Key Takeaways & Best Practices:

- Always evaluate your retrieval engine with synthetic Q&A datasets.
- Apply chunk overlap and metadata filtering by user tenant.
- Use context compression to save token costs and reduce noise.

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Chapter 2: Autonomous Agent Loops and Safe Tool Execution

Deterministic orchestration and sandbox boundaries

LLM agents must execute actions through strictly typed schema declarations. By combining tool calling with schema validation, agent errors are caught prior to execution.

Key Takeaways & Best Practices:

- Enforce timeouts and max iteration bounds on all autonomous agent loops.
- Audit all outgoing tool parameters before mutating databases.